ELECTRONICS & ROBOTICS CLUB
IIT Bombay

erciitbombay@gmail.com
erc.tech-iitb.org
+91 924 418 1630 | +91 966 085 2364

Session-1 Mechatronics Assignment

Submission Deadline: 11:59 PM, 9th June 2026

Submit your solutions (handwritten or typed) in pdf format and submit the drive link

---

SITUATION

A medical robotics startup has spent three years building **AESOP-7**, a 3-DOF robotic instrument arm designed to assist in minimally invasive surgery.

Today is the final day of pre-clinical validation. The robot must pass a series of precision positioning tests on a synthetic tissue phantom before regulatory submission next week.

Midway through Test Suite 4, the calibration module throws an unhandled exception and crashes. Position tracking is gone. The arm is physically fine - motors work, joints move but the software no longer knows where the instrument tip is relative to the camera, the phantom, or its own base.You have to fix it.

Arm specs throughout: L1 = 0.35 m, L2 = 0.25 m, L3 = 0.15 m

Q1. The Arm Went the Wrong Way

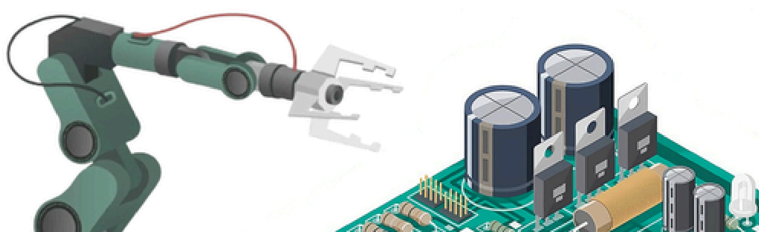
During Test Suite 3, AESOP-7 was commanded to position the instrument tip at phantom target **T1: (0.28 m, 0.32 m)** in the arm base frame. The IK solver returned:

$\theta_1 = 38.7°$, $\theta_2 = 57.3°$

The arm moved. The instrument tip arrived at **(0.28 m, −0.32 m)** ,a perfect reflection across the X-axis. The test logged a failure but nobody caught it before Suite 4 began.

a) Without running any new calculation, identify the **single specific term** in the rotation matrix responsible. What does that term control physically, and what happens to every Y-coordinate in the system when it flips?

Institute Technical Council
Students Gymkhana
IIT Bombay, Powai
Bombay 76



b) The corrupted software is using this rotation matrix:

$$R(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$$

Identify **two distinct errors** : one is a sign error on a specific entry, the other is a fundamental mathematical property that all valid rotation matrices must satisfy. State the property and show that this matrix violates it.

c) The pre-validation bench test used calibration targets placed only along the arm's centreline (y ≈ 0). Explain precisely why this test geometry failed to catch the Y-reflection bug. What is the minimum change to the test configuration that would have revealed it?

d) The same corrupted matrix is active in the camera-to-base transform. The camera sees target T1 at **(0.22 m, 0.18 m)** in the camera frame, with a **−15° rotation** between camera and base. Compute where the arm actually moves versus where it should go. State the positional error in centimetres.

Q2. The Misconception in the Codebase

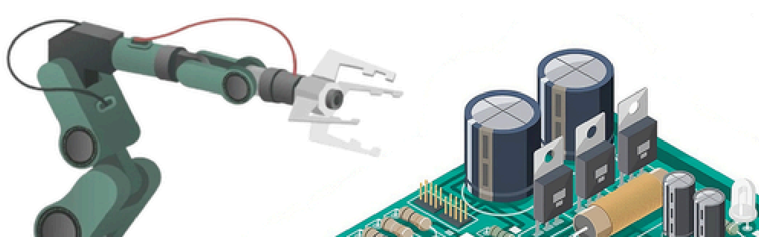
Searching through the version history you find this comment, written eight months ago:
"AESOP-7's transform chain uses homogeneous matrices. Since rotation matrices are orthogonal, R-inverse = R-transpose. Therefore to convert any point from instrument frame back to base frame, you simply transpose the entire 4x4 homogeneous matrix T. This is computationally efficient and mathematically equivalent to a full matrix inverse."

This contains one correct statement, one partially correct statement, and one statement that is simply wrong.

a) Identify all three. Quote the relevant phrase and explain precisely what is right, partially right, or wrong about each.

b) Derive the correct form of **T^{-1}. Start from $T \times T^{-1} = I$** Partition both matrices into their R and t blocks, and solve for each block algebraically. Show every step.

c) If the codebase transposes the full **T** rather than inverting correctly, describe the physical consequence during a positioning test -not just 'it goes to the wrong place' but specifically what the test log would record versus what actually happened, and why this discrepancy could pass unnoticed across multiple test runs.



Q3. Three Broken Prototypes

During development, three earlier control system architectures were tested and rejected. For each one: name the failure mode precisely, describe a realistic scenario where it causes the positioning test to fail, and propose the minimum fix.

System X
The arm moves to the IK-computed position. The instrument contacts the phantom. No contact force is read during motion —-force sensing activates only after joint encoders confirm target angles have been reached. The system logs success when encoder targets are met.
System Y
Force sensors on the instrument are active throughout. When contact force exceeds 1.2 N the arm halts and logs contact. However the camera feed is paused during arm movement to reduce processing load.
System Z
Camera streams continuously. Force feedback is active. IK updates at 20 Hz. However the joint encoders on all three joints have a systematic offset of +3° - they consistently report angles 3° higher than actual, identically on every joint, without drift

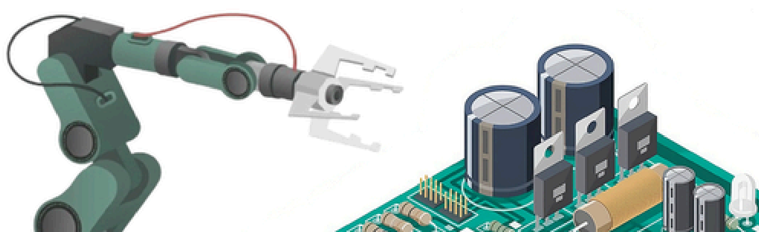
---

Q4. Explain It To The Project Manager

The project manager is a sharp, non-technical person .Write for her specifically. No equations. No jargon without immediate plain-English explanation.

a) **Gimbal lock**
What degree of freedom is lost, why rotating further cannot recover it, and why AESOP-7 uses quaternions for instrument orientation tracking instead of Euler angles. Use any analogy drawn from everyday office or business life.

b) **Workspace singularities**
What they are, why they matter for a positioning test specifically, and what the test operator should do if the arm starts behaving erratically near the edge of its range.



erciitbombay@gmail.com
erc.tech-iitb.org
+91 924 418 1630 | +91 966 085 2364

Q5. Design the Contact Verification System

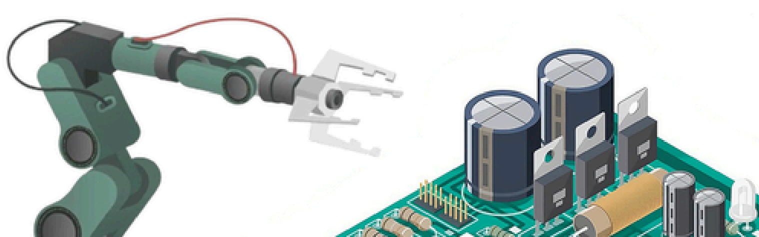
AESOP-7 has moved to target T1 on the phantom. The instrument tip has stopped.
How do you confirm the tip is in contact with the correct target marker and not an adjacent one?
a) Specify what **sensors** you would use and exactly what each one measures. You may use what is already on AESOP-7 (camera, force sensors, joint encoders) or propose additions but every addition must be justified against the constraint that the validation budget allows no hardware changes to the arm itself. External additions to the test rig are acceptable.
b) Identify two ways your verification system could produce a **false positive** -logging a pass when the instrument has not correctly contacted the target. For each, describe the exact physical situation and what the test log would incorrectly show.

Q6. The Orientation-Constrained Target
Test Suite 4 requires the instrument to reach phantom target T2 at **(0.30 m, 0.38 m)** with instrument orientation **$\varphi = -25°$**
$\varphi = \theta 1 + \theta 2 + \theta 3$

a) The wrist link L3 points at angle phi. Subtract its contribution from the target to get the corrected 2-DOF IK target. Then solve for both theta2 values.

b) For each θ2 solution compute the required theta3 from the orientation constraint.

c) One solution requires $|\theta 3| > 135°$. The mechanical joint limit on theta3 is ±130°. Identify which solution this is. Then answer: if both solutions violated the joint limit, what options does the engineer have? Consider whether the target position, the approach angle, or the robot mounting position could be adjusted, and what the test implications of each are.



ELECTRONICS & ROBOTICS CLUB
IIT Bombay
erciitbombay@gmail.com
erc.tech-iitb.org
+91 924 418 1630 | +91 966 085 2364

Q7. The Transform Chain -Rebuilt

AESOP-7 is in this configuration: **θ1 = 45°, θ2 = −60°, θ3 = 30°**
Single-joint homogeneous transform:

$$T(\theta, L) = \begin{bmatrix} \cos\theta & -\sin\theta & L\cos\theta \\ \sin\theta & \cos\theta & L\sin\theta \\ 0 & 0 & 1 \end{bmatrix}$$

a) The top-right column of (T1 × T2) gives the wrist joint position in the base frame. Without multiplying the full matrices, explain why -trace what happens to the translation block through the multiplication.

b) Compute the instrument tip position using FK equations. Show all cumulative angles and intermediate terms.

c) Midway through the test, the technician adjusts the test rig mounting and the entire robot base tilts 10°. Joint angles are unchanged. Compute the new instrument tip position in the room frame. State clearly whether you need to re-run IK and why.

Q8. IK — Two Targets, Full Consequences

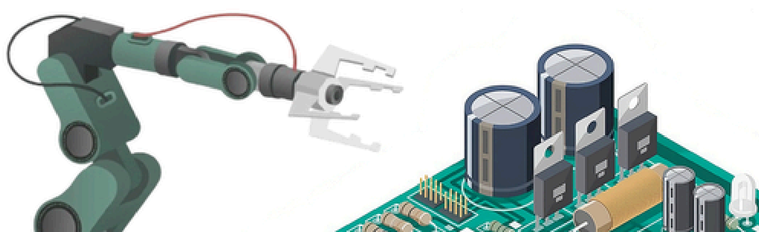
Two targets must be reached in Test Suite 4:

T1 - Precision contact test: (0.30 m, 0.38 m)
T2 - Lateral reach test: (−0.20 m, 0.52 m)

IK formulas (2-DOF reduction, L3 wrist fixed):

$\cos(\theta_2) = (x^2 + y^2 - L_1^2 - L_2^2) / (2 \cdot L_1 \cdot L_2)$
$\theta_2 = \text{atan2}(\pm\sqrt{1 - \cos^2(\theta_2)}, \cos(\theta_2))$
$\theta_1 = \text{atan2}(y, x) - \text{atan2}(L_2 \cdot \sin(\theta_2), L_1 + L_2 \cdot \cos(\theta_2))$

For **each target**:
a) Compute cos(θ2) and both θ2 solutions
b) Compute both θ1 solutions



Institute Technical Council
Students Gymkhana
IIT Bombay, Powai
Bombay 76

ELECTRONICS & ROBOTICS CLUB
IIT Bombay
erciitbombay@gmail.com
erc.tech-iitb.org
+91 924 418 1630 | +91 966 085 2364

c) T2 - **physical obstruction check**: The test rig frame occupies the region **$x < -0.08$ m** AND **$y < 0.38$ m** in base frame. The arm must not pass through this region at any point during its motion. Using your computed angles, reason geometrically about which IK solution risks passing through the rig frame. You do not need to trace every intermediate point -reason from the overall arm shape each solution produces.

d) **The deeper question:** Both IK solutions for T1 are within joint limits and clear of obstacles. The team always defaults to elbow-down. Describe a specific physical scenario -a particular test rig attachment or cable routing - where elbow-down causes a collision that elbow-up avoids. Be specific about the geometry.

Institute Technical Council
Students Gymkhana
IIT Bombay, Powai
Bombay 76